Stochastic FRAMES: experiment specification

Nonconvex ℓ_1-constrained least squares in dimension 1024

14 October 2026 · Random negative-vertex initialization

Problem. Let $d = 1024$, and let e_j denote the vector with entry 1 in position j and 0 elsewhere. The problem is

$$\boxed{\min_{x\in C} F(x), \qquad F(x) = f(x) + g(Tx).}$$

The four ingredients are

$$\begin{aligned} f(x) &= \frac{1}{4}\big(\|x-a-b\|_2^2 + \|x-a+b\|_2^2\big) - \frac{5}{8}(u^\top x)^2 = \frac{1}{2}\|x-a\|_2^2 + \frac{1}{8} - \frac{5}{8}(u^\top x)^2, \\ C &= \{x \in \mathbb{R}^{1024} : \|x\|_1 \le 1\}, \qquad T = I_{1024}, \\ g(z) &= \iota_D(z) = \begin{cases} 0, & z \in D, \\ +\infty, & z \notin D, \end{cases} \qquad D = \mathbb{R}^{1024}_+. \end{aligned}$$

Thus original feasibility means $x \ge 0$ and $\sum_j x_j \le 1$. The data vectors are

$$a_j = \begin{cases} 1/32, & 1 \le j \le 64, \\ -3/128, & 65 \le j \le 128, \\ 1/128, & 129 \le j \le 1024, \end{cases} \qquad b_j = \frac{(-1)^{j-1}}{64}, \qquad u_j = \begin{cases} (-1)^{j-1}/8, & 1 \le j \le 64, \\ 0, & 65 \le j \le 1024. \end{cases}$$

With $Q := I - \frac{5}{4}uu^\top$, the true smooth gradient is $\nabla f(x) = Qx - a$. Since $\|u\|_2 = 1$, the Hessian Q has eigenvalues $-1/4$ and 1: the objective is nonconvex and globally 1-smooth. Nonconvexity also occurs within the feasible set: if x_s has its first 64 entries equal to $1/64$ and all other entries zero, then

$$x_s + tu \in C \cap D, \qquad F(x_s + tu) = \frac{91}{512} - \frac{t^2}{8}, \qquad |t| \le \frac{1}{8}.$$

Finite sum and stochastic oracle. Sample one of two components uniformly:

$$f_\varepsilon(x) = \frac{1}{2}\|x - a + \varepsilon b\|_2^2 - \frac{5}{8}(u^\top x)^2, \qquad \varepsilon \in \{-1, +1\}, \qquad f = \tfrac{1}{2}(f_{-1} + f_{+1}).$$

At iteration k, draw a fresh independent ε_k and use $q_k = \nabla f_{\varepsilon_k}(x_k) = Qx_k - a + \varepsilon_k b$. Writing $\mathcal{F}_k$ for the history before this draw (including x_0),

$$\mathbb{E}[q_k \mid \mathcal{F}_k] = \nabla f(x_k), \qquad \mathbb{E}[\|q_k - \nabla f(x_k)\|_2^2 \mid \mathcal{F}_k] = \|b\|_2^2 = \frac{1}{4}.$$

This is one sampled least-squares *block* gradient per update, with the common concave quadratic included.

Initialization and run parameters. For each seed $s \in \{0, \dots, 7\}$, independently of the oracle draws, sample J_s uniformly from $\{1, \dots, 1024\}$ and set

$$\boxed{x_0^{(s)} = -e_{J_s}.}$$

The initial point has one randomly chosen entry equal to -1 and all other entries equal to 0. It satisfies $\left\|x_0^{(s)}\right\|_1 = 1$ and $\mathrm{dist}_D(x_0^{(s)}) = 1$. The initial momentum estimate is the first sampled gradient: $m_0 = q_0$.

The experiment uses $N = 32768$ updates per seed, 8 seeds, and float64 arithmetic. For $k = 0, \dots, N-1$, the schedules are

$$\boxed{\gamma_k = (k+1)^{-3/4}, \qquad \beta_k = (k+1)^{-1/4}, \qquad \alpha_k = (k+1)^{-1/2}.}$$

Here γ_k is the Frank–Wolfe step size, β_k the smoothing parameter, and α_k the weight on the new stochastic gradient. All prefactors are 1; there is no warm-up or schedule shift.

Gap and feasibility notation. Define $(x_k)_- = \min(x_k, 0)$ coordinatewise, and set

$$z_k := \nabla f(x_k) = Qx_k - a, \qquad h_k := z_k + \frac{(x_k)_-}{\beta_k}.$$

The reported gaps use these exact vectors. Their definitions and explicit formulas are

$$G_k := \max_{v \in C} \langle h_k, x_k - v \rangle = \boxed{\langle h_k, x_k \rangle + \|h_k\|_\infty}, \qquad \text{smoothed Frank–Wolfe gap,}$$

$$H_k := \max_{v \in C \cap D} \langle z_k, x_k - v \rangle = \boxed{\langle z_k, x_k \rangle + \max\{0, -\min_j z_{k,j}\}}, \qquad \text{original constrained gap,}$$

$$\delta_k := \mathrm{dist}_D(x_k) = \boxed{\|(x_k)_-\|_2}, \qquad \text{feasibility violation.}$$

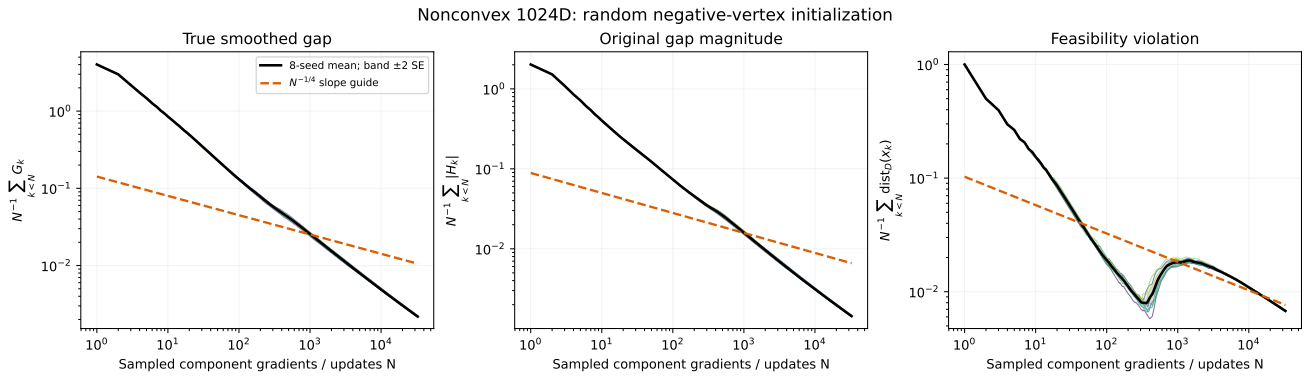
Since $x_k \in C$, $G_k \geq 0$. However, H_k can be negative when $x_k \notin D$; the original-gap plot therefore reports $|H_k|$. Feasibility and the original gap are interpreted together.

What the plots average. For $Z_k \in \{G_k, |H_k|, \delta_k\}$, define

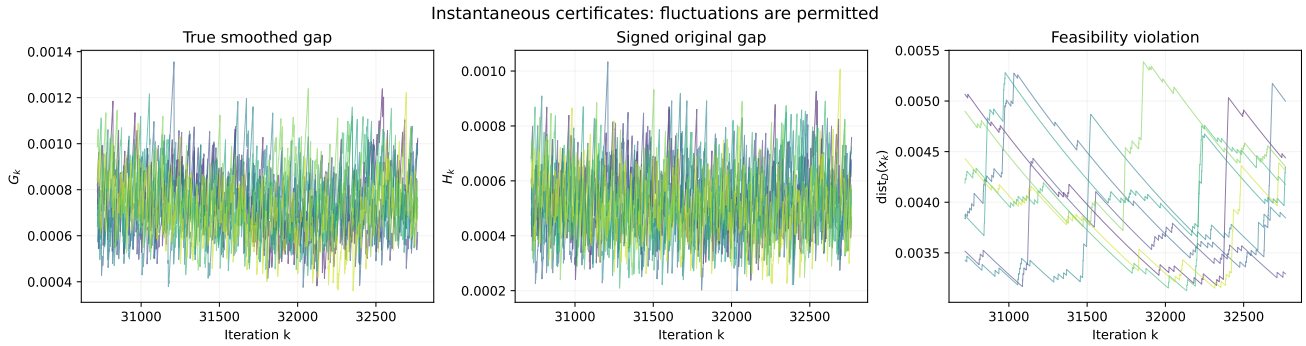
$$\overline{Z}_N^{(s)} = \frac{1}{N} \sum_{k=0}^{N-1} Z_k^{(s)}, \qquad \widehat{A}_Z(N) = \frac{1}{8} \sum_{s=0}^{7} \overline{Z}_N^{(s)}.$$

Every pre-update iterate $x_0, \ldots, x_{N-1}$ is included. Thin curves are individual seed averages; the black curve is $\widehat{A}_Z(N)$; the band is ± 2 standard errors across the eight seed averages (descriptive). The dashed $N^{-1/4}$ line is a slope guide, not a numerical upper bound.

Plots. Eight seeds with random negative-vertex initialization.



Cumulative averages of G_k, $|H_k|$, and δ_k over all recorded iterates.



Individual values of G_k, signed H_k, and δ_k during the final 2048 iterations. Each color represents one seed.